

Nicholas Hu

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EDUCATION

Rice University, Houston, TX

May 2027

BS in Electrical and Computer Engineering, Minor in Data Science

GPA: 3.65/4.0

Honors: National Merit Scholar (Rockwell Automation) | Chevron Engineering Scholarship (sole ECE recipient)

Relevant Coursework: Data Science Tools and Models, Artificial Intelligence, Machine Learning Concepts, Implementation of Digital Systems, Introduction to Database Systems

EXPERIENCE

Rockwell Automation

Milwaukee, WI

Software Development Intern

June 2026 – August 2026

- Redesigned duplicate slot-address validation in Kotlin, replacing one bus-level warning with per-device errors linked to each conflicting profile so users can identify and correct the issue.
- Corrected a topology lookup defect that could return the first topology instead of the device's actual topology; added **3 unit tests**, updated **4 browser-based end-to-end cases**, and merged the branch after CI and SonarQube checks passed.
- Took ownership of a senior developer's in-progress performance optimization during their absence, learned the existing implementation, and delivered it **4 days ahead of schedule**.

Software Development Intern

June 2025 – August 2025

- Automated **3 recurring manual regression cases** in TypeScript, published the browser tests through Jenkins, and validated successful execution across qTest runs.
- Maintained software quality across a **22-test nightly regression suite** by repeatedly tracing failed end-to-end tests to recent merge requests and recommending reverts of the responsible changes to developers before defects could reach users.

Control and Firmware Engineering Intern

June 2024 – August 2024

- Built a MATLAB/Simulink workflow that added visual model comparisons to GitLab, reducing developers' manual comparison work and earning adoption by **2 additional engineering teams**.
- Evaluated drive-control software across **200 automated tests** and established a performance baseline through **32 system efficiency tests** spanning motor loads and speeds.

Rice University ECE & Baylor College of Medicine

Houston, TX

ML Student Researcher

November 2024 – Present

- Built a Python/XGBoost pipeline using wearable and smartphone features from a **26-participant, 90-day depression study**; used leave-one-subject-out cross-validation to prevent participant leakage and achieved **0.74 AUC** at baseline.
- Analyzed clinical, quality-of-life, and wearable measures to identify a statistically significant relationship between limb-threat severity and toe-brachial index; coauthored a **Journal of Vascular Surgery** publication from the Rice-Baylor collaboration.

PROJECTS

GoldBench (Quantitative Forecasting Benchmark) | *Python, LightGBM, PyTorch, ONNX*

May 2026

- Built a reproducible forecasting benchmark comparing **5 model families** across **5 years of gold-futures data** and **28 time-ordered evaluation periods**, finding that market-direction predictions remained near chance while volatility showed a stronger predictive signal.
- Converted a LightGBM model to ONNX, reducing model size by **34%** and accelerating inference by **2.8 times** while preserving identical predictions.

FPGA Peripheral System | *SystemVerilog, Python, AXI4-Lite, Vivado, cocotb, Verilator, SystemRDL*

April 2026

- Designed a Cmod A7-35T FPGA system that lets a host control GPIO, PWM, and timer peripherals through a UART to AXI4-Lite bridge and asynchronous FIFO spanning **2 clock domains**.
- Resolved failing UART tests by tracing width truncation before subtraction and correcting both the receive and transmit paths.
- Defined **14 registers** in SystemRDL to generate FPGA logic and Python test mappings; passed cocotb/Verilator simulations, **6/6 hardware-in-the-loop tests**, and Vivado timing analysis up to **153.6 MHz**.

LEADERSHIP

OwlSat (Rice CubeSat Club) | *Hardware Team Lead*

September 2023 – Present

- Lead a cross-functional **7-member hardware/software team** integrating CubeSat subsystems under size, weight, and power constraints.
- Restored CubeSat function during hardware bring-up by tracing a suspected component failure to an incorrectly labeled part and identifying the correct component.

Rice IEEE | *Class Representative*

August 2023 – Present

- Led senior-class outreach and event operations for the club's first professional-development speaker series, bringing engineers and recruiters from Samsung, HPE, and Y Combinator to campus and reaching **63 students (97% of the ECE graduating class)**.

SKILLS

Languages: Python, Java, Kotlin, TypeScript, C/C++, SystemVerilog, SQL, MATLAB, Bash

Development & Tools: Jenkins, SonarQube, Docker, SQLite, pandas, SDLC, CI/CD, Agile/Scrum

Certifications: Lean Six Sigma Yellow Belt | Advanced Generative AI for Developers (Google Cloud) | Getting Started with Deep Learning (NVIDIA)